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Original Article

Real-time PCR for the rapid detection of *vanA*, *vanB* and *vanM* genes

Yuan-Hui He [a,1], Gen-Jie Ruan [a,1], Hui Hao [b], Feng Xue [a], Ya-Kun Ma [a], Sai-Nan Zhu [c], Bo Zheng [a,*]

[a] *Institute of Clinical Pharmacology, Peking University First Hospital, Beijing, China*
[b] *IPE Biotechnology Co. Ltd, Beijing, China*
[c] *Department of Biostatistics, Peking University First Hospital, Beijing, China*

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Abstract *Background/Purpose:* To evaluate the ability of quadruple Taqman probe real-time PCR to the detection of *vanA*, *vanB* and *vanM* in enterococcal isolates.
Methods: A total of 343 strains, including 253 vancomycin-resistant *enterococcus* (VRE) strains and 90 non-VRE strains, were tested by both quadruple Taqman probe real-time PCR and gel-based PCR assay.
Results: When differentiating among three genotypes of *vanA*, *vanB* and *vanM* in VRE strains, the sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), diagnostic accuracy and consistency of the quadruple Taqman probe real-time PCR were all 100%. Minimum.

Inhibitory concentration (MIC) results showed that there was a wide MIC range of vancomycin and teicoplanin for the strains that harboring *vanA*/*vanM* gene respectively or harboring *vanA* and *vanM* genes simultaneously. However, the VRE strains with *vanB* genotype all were sensitive to teicoplanin.
Conclusion: Considering the excellent PPV and low NPV, real-time PCR would be useful to monitor VRE-colonized or infected patients. However, further evaluation of the assay's performance in the clinical specimens is required, especially when considering that high level of PCR inhibitors present in these samples.
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* Corresponding author. Institute of Clinical Pharmacology, Peking University First Hospital, No. 8, Xishiku Street, Xicheng District, Beijing, 100034, China. Fax: +8610 62072817.
E-mail address: doctorzhengbo@163.com (B. Zheng).
[1] Equal contribution.

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Introduction

First isolated in the late 1980s, vancomycin-resistant enterococci spread rapidly throughout the United States, Europe and elsewhere.[1] VRE can increase treatment costs, prolong hospital stay, and lead to high morbidity, which is associated with bloodstream infection.[2] In China, the resistance genotype mainly consists of *vanA*, *vanB* and *vanM* genotype. *vanA* genotype, the major resistant genotype of VRE, shows that enterococci were resistant to vancomycin and teicoplanin.[3] *vanM* type also shows a high resistance to vancomycin and teicoplanin.[4–6]

Active screening and rapid test of the genotype of enterococci is considered as effective method to reduce the incidence of VRE-related diseases.[3] Recently, Nomura and his collogues have developed a new method to detect seven genes simultaneously including *vanA*, *vanB* and *vanM* gene.[7] However, compared to multiplex PCR method, multiplex real-time PCR method obtained the results more rapidly. At present, multiplex real-time PCR methods are mostly applied for *vanA* type and *vanB* type.[8–13] However, it lack of the ability to detect *vanA*, *vanB* and *vanM* gene simultaneously. In this study, we evaluated the performance of quadruple real-time PCR to detect *vanA*, *vanB* and *vanM* gene.

Methods and materials

Bacterial strains

A total of 343 strains, including 253 vancomycin-resistant *Enterococcus faecium* strains and 90 non-VRE strains, were collected by Institute of clinical pharmacology, Peking University First Hospital. Among the 90 non-VRE strains, there were 20 vancomycin-sensitive *Enterococcus faecalis* isolates, 20 *Enterococcus faecium* isolates, 10 *Staphylococcus aureus* isolates, 10 *Klebsiella pneumoniae* isolates, 10 *Escherichia coli* isolates, 10 *Pseudomonas aeruginosa* isolates and 10 *Acinetobacter baumannii* isolates. *E. faecium* ATCC29212 and *E. faecium* BM4147 were used as the control strains.

Agar culture method and MICs test

Bacterial strains were inoculated to BHA medium after resuscitation, isolated to single colony with plate line method, and then incubated at 37 °C for 24 hours. MICs of vancomycin and teicoplanin were determined by agar dilution method according to CLSI. The breakpoints were also described in the CLSI.[14]

DNA extraction

We picked single colony into 200 μL sterile saline and boil it for 10 min at 100 °C. After boiling, the liquid contained bacteria was centrifuged for 30sec with 12000r/sec. We then took the supernatant as the template for PCR and quadruple real-time PCR.

Quadruple real-time PCR method

The procedure was carried out according to the instruction from the quadruple Taqman probe real-time PCR kit (2010806, IPE, Beijing, China). The reaction system includes reaction liquid which consists of deoxyribonucleoside triphosphate (dNTP), the primers and probes of *vanA*, *vanB*, *vanM* and internal control and reaction enzyme which consists of Deoxyribonucleic acid polymerase and Uracil glycosylase. Within the 25.5 μL reaction system, there are 20 μL reaction solution, 0.5 μL reaction enzyme, 5 μL template. The sequence of primer and probe of Taqman real-time PCR method are listed in Table 1. The following PCR program was performed on the LightCycler (LC; Idaho Technology Inc., Idaho Falls, Idaho). Cycling conditions were as follows: initial denaturation at 50 °C for 2 min, 95 °C with a 5 min hold, followed by 40 cycles of 15 sec at 95 °C, 60 sec at 60 °C.

PCR gel-based method

The 20 μL reaction system is consisted of 1 μL upstream primer (10 μmol/L), 1 μL downstream primer (10 μmol/L), 10 μL 2 of Taq PCR MasterMix (KT201, TIANGEN, Beijing, China), 6 μL sterile ultrapure water and 2 μL DNA templates. *VanA*, *vanB*, *vanM* and van-N primer and other reaction sequences and reaction procedures were described in Nomura's work.[7] Each batch of reactions contained a negative control, a positive control and a blank control, which are identical to the quadruple Taqman probe real-time PCR method.

Table 1 Details of the primer and reaction condition in quadruple real-time PCR method.

Target gene	Primer	Sequence (5′-3′)	Channel	Reference
vanA	A1	GCCGGAAAAAGGCTCTGAA	FAM	This study
	A2	TTTTTTGCCGTTTCCTGTATCC		
	probe	FAM-CGCAGTTATAACCGTTCCCGCAGACC-BHQ1		
vanB	B1	GATTTGATTGTCGGCGAAGTG	JOE/VIC	This study
	B2	TCCTGATGGATGCGGAAGA		
	probe	VIC-TCAAATCCGGCTGAGCCACGGT-BHQ1		
vanM	M1	GAAAAAGGTTCTGAAAACGCAACTA	ROX	This study
	M2	TTTTTTGCCGCTTCTTTTATCC		
	probe	ROX-AACCGTTCCCGCAGAACTATCGGC-BHQ1		
Internal control	E-16S-F2	GCGGTGGAGCATGTGGTT	Cy5	This study
	E-16S-R2	CTCTAGAGTGGTCAAAGGATGTCAAG		
	E-16S-P2	Cy5-ATTCGAAGCAACGCGAAGAACCTTACCA-BHQ3		

Table 2 The detection details of quadruple real-time PCR when choosing PCR gel-based method as reference.

Gene type	No. of isolates with each set of results[a]				(%)					Kappa coefficient (*P*)
	Reference +		Reference -		Sensitivity	Specificity	PPV	NPV	Diagnostic Accuracy	
	test +	test -	test +	test -						
vanA	213	0	0	130	100 (98.28, 100[b])	100 (97.20, 100[b])	100 (98.28, 100[b])	100 (97.20, 100[b])	100 (98.93,100[b])	1 (<0.001)**
vanB	5	0	0	338	100 (47.82, 100[b])	100 (98.91, 100[b])	100 (47.28, 100[b])	100 (98.91, 100[b])	100 (98.93, 100[b])	1 (<0.001) **
vanM	15	0	0	328	100 (78.20, 100[b])	100 (98.88, 100[b])	100 (78.20, 100[b])	100 (98.88, 100[b])	100 (98.93, 100[b])	1 (<0.001) **
vanA& *vanM*	20	0	0	323	100 (83.16, 100[b])	100 (98.86, 100[b])	100 (83.16, 100[b])	100 (98.86, 100[b])	100 (98.93, 100[b])	1 (<0.001) **

[a] Reference: when choosing PCR gel-based method as reference; reference+, when the result of PCR gel-based method is positive; test+, when the result of quadruple real-time PCR method is positive.

[b] 95% confidence interval.

** the consistency is good; PPV = positive predictive value; NPV = negative predictive value; *vanA* = The strains is only harboring *vanA*; *vanB* = The strain is only harboring *vanB*; *vanM* = The strain is only harboring *vanM*; *vanA* & *vanM* = The strain is harboring *vanA* and *vanM* gene simultaneously.

Statistical analysis

95% confidence interval of sensitivity, specificity, PPV, NPV, diagnostic accuracy of quadruple real-time PCR method is calculated. McNemar test is used to analyze the statistical difference of sensitivity and specificity among the experiment method and the reference methods. The consistency of the results of quadruple real-time PCR method and multiplex PCR gel-based method is validated by Kappa test method: if Kappa coefficient $\geq$0.75, the consistency is good; if Kappa coefficient <0.75, $\geq$0.40, the consistency is general; if Kappa coefficient <0.40, the consistency is not ideal. $P < 0.05$ (two-tailed) showed statistical significance. The statistical software used in this study is GraphPad Prism 5.0 software (GraphPad Software, Inc., La Jolla, CA).

Ethical approval

Ethics committee approval is not required as the patients are de-identified.

Results

Both quadruple real-time PCR method and multiplex PCR gel-based method accurately distinguished 253 VREs and 90 non-VRE strains. There were 213 strains of *vanA* type VRE, 5 strains of *vanB* type VRE, 15 strains of *vanM* type VRE and 20 strains of VRE containing both *vanA* and *vanM* gene. Both test methods showed negativity for all 90 non-VRE strains.

Sensitivity, specificity, PPV, NPV, diagnostic accuracy and consistency

When choosing PCR gel-based method as the reference, our study shows that the sensitivity, specificity, PPV, NPV, diagnostic accuracy and consistency of quadruple real-time PCR to detect *vanA* gene, *vanB* gene, *vanM* gene, *vanA* and *vanM* gene simultaneously in the same strain were all 100%. There is no statistical difference between the two methods ($P = 1$, >0.05). The consistency between this two methods are good ($P < 0.001$) (Table 2).

MIC results

Vancomycin has a wide MICs distribution of enterococci with *vanA*, *vanB*, *vanM* and *vanA* & *vanM* gene; Teicoplanin has a MIC of 1 μg/mL for strains carrying *vanB* gene and a wide range of MICs for carrying *vanA*, *vanM* and *vanA* & *vanM* Enterococcus (Table 3).

Table 3 MIC range of different genotype VRE to vancomycin and teicoplanin.

Genotype (number of isolates)	Vancomycin (μg/mL)	Teicoplanin (μg/mL)
vanA (213)	32—512	1—256
vanB (5)	32—256	1
vanM (15)	16—512	1—256
vanA & *vanM* (20)	64—512	8—128

Detection time

The time to acquire the results by PCR gel-based method and quadruple real-time PCR method were about 4 hours and 1.6 hours, respectively. PCR gel-based method includes PCR and DNA cleavage agarose gel electrophoresis. The time required for the PCR process is about 2 hours (the details are described in Nomura's study).[7] The time required for agarose gel preparation is about 40 min. The time for electrophoresis is about 50 min. It takes 15—20 min per sample to configure the PCR reaction mixture and read electrophoresis result on average. The details of reaction procedures of the Real-time PCR method including 50 °C for 2 min, 95 °C with 5 min hold and followed by 40 cycles of amplification at 95 °C with a 15-second hold, 60 °C with a 1 min, reading fluorescence. It takes about 1.5 hours to obtain fluorescence reaction results, while the procession of configuring reactants and result reading takes 5—10 min on average for each sample.

Discussion

Several previous studies were conducted to evaluate the effects of molecular biology methods to detect *vanA* and *vanB* VRE. However, there is no report of methods for detecting *vanA, vanB, vanM* VRE simultaneously except recently Normura reported.[7] In this study, *vanA*, *vanB* and *vanM* VRE were detected by real-time PCR with quadruple Taqman probe. The sensitivity, specificity, PPV and NPV were satisfactory. It can directly detect the resistance gene and avoid missed diagnosis caused by phenotypic detection of Vancomycin-resistant variable enterococci.

vanA has been the most common vancomycin-resistant enterococci genotype in China[9,12,15—19]; In recent years, *vanM* gene has become a more common genotype in Shanghai, Hangzhou city and so on[4,20]; Recently, we have found that some enterococci harboring *vanM* gene which are sensitive to both vancomycin and teicoplanin by agar dilution method expressed heterogeneously resistant to vanmcomycin. These strains cannot be isolated by conventional screening methods. Therefore, the detection of *vanA*, *vanB* and *vanM* genes by real-time fluorescent quantitative PCR is beneficial to the rapid detection of these resistance genes in Enterococcus.

Although the cost of the method is higher than that of PCR gel-based method, it avoids the cumbersome and time-consuming preparation procedure of the PCR gel-based method. Therefore, it is expected to be the method to detect VRE in clinical samples, which is conducive to take infection control interventions in time,[21] to provide timely and reasonable medication information,[22] and to exclude VRE carriers rapidly.[19] To further validate this method, more clinical samples will be needed in the future.

Conclusion

The molecular detection method is fast and accurate, which avoids the time-consuming and tedious preparation procedures in the PCR gel-based method, but it needs to be further verified in clinical isolates. It is expected to be the method that can detect the clinical isolates in time and obtain the test results effectively.

Conflicts of interest

The authors have declared no conflict of interest.

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